

Problem 4.5

What simple interest rate is necessary for \$10,000 to earn \$100 interest in 15 months?

Solution

For simple interest,

$$I = P(i)(t).$$

Here,

$$I = 100, \quad P = 10000, \quad t = 15 \text{ months} = 1.25 \text{ years}.$$

Thus,

$$100 = 10000(i)(1.25).$$

So,

$$i = \frac{100}{10000 \cdot 1.25}.$$

Therefore,

$$i = 0.008.$$

Hence,

$$i = 0.8\%.$$